BUSS386 Midterm Exam

Futures and Options

Thursday, 21 May 2026 — 15:10–17:10

Name: _____ ID: _____

Format. 2 hours. Closed book. One A4 hand-written cheatsheet (both sides) allowed; typed sheets not permitted. Total: 100 points.
Coverage. Lectures 1–4 (intro to derivatives, interest rates, forwards and futures, forward/futures pricing).

Section A — True / False (10 × 2 = 20 pts)

For each statement, write **T** or **F**.

A1. The no-arbitrage forward price of a non-dividend-paying stock is always greater than the current spot price, assuming a positive risk-free rate.

A2. The dollar duration of a bond is defined as its modified duration times its current price.

A3. In a cross hedge, the minimum-variance hedge ratio must always be less than 1.

A4. Holding maturity, coupon, and yield level constant, a bond with a higher coupon rate has a *longer* Macaulay duration.

A5. KOSPI 200 futures are physically settled at expiry, requiring delivery of a basket of 200 stocks.

A6. If interest rates are constant or deterministic, the futures price equals the forward price for the same underlying and maturity.

A7. SOFR is computed daily from actual overnight Treasury repurchase transactions, not from a survey of banks.

A8. The implied forward rate from today's yield curve is the market's unbiased expectation of the future spot rate.

A9. A 4% nominal rate per annum compounded continuously is exactly equivalent to a 4.0811% effective annual rate.

A10. At the maturity of a futures contract, the basis (spot minus futures) converges to zero.

Section B — Multiple Choice (10 × 3 = 30 pts)

Circle the single best answer.

B1. Which best describes the role of a clearing house (CCP) for futures?

(a) Sets daily settlement prices for OTC trades.

(b) Becomes the counterparty to every cleared trade, eliminating bilateral credit risk.

(c) Makes markets in OTC derivatives.

(d) Regulates margin requirements at the exchange level.

B2. A 1-day 99% VaR for a \$5M position with daily volatility of 1.5%, assuming normally distributed returns, is closest to: ($\Pr(z \leq 0.01) = z_{0.01} = 2.326$)

(a) \$123,750

(b) \$147,000

(c) \$174,450

(d) \$232,600

B3. The Macaulay duration of a 3-year, 6% annual-coupon bond priced at par is closest to:

(a) 2.50 years

(b) 2.83 years

(c) 3.00 years

(d) 3.17 years

B4. A stock pays a known dividend of \$1 in 6 months. The current stock price is \$50, $r = 4\%$ (c.c.), and the yield curve is flat. The no-arbitrage 1-year forward price is closest to:

(a) \$49.02

(b) \$50.00

(c) \$51.02

(d) \$52.04

B5. An airline that will need to buy 1,000,000 gallons of jet fuel in 6 months wants to hedge price risk with futures. The appropriate hedge is:

(a) Short jet-fuel futures (sell forward).

(b) Long jet-fuel futures (buy forward).

(c) Buy a put option on jet fuel.

(d) No hedge is needed because the airline always passes costs through.

B6. The 6-month forward price for an equity index with $S_0 = 500$, $r = 4\%$, dividend yield $q = 1.5\%$ (both c.c.) is closest to:

(a) 493.83

(b) 503.13

(c) 506.29

(d) 510.05

B7. A bank will receive \$5M in 3 months and plans to deposit it for 3 months. To hedge against falling deposit rates between now and then, the appropriate SOFR-futures position is:

(a) Short SOFR futures.

(b) Long SOFR futures.

(c) No futures needed.

(d) Use FRAs only, never futures.

B8. The 1-year zero rate is 3% and the 2-year zero rate is 4% (both c.c.). The implied 1-year-forward rate starting one year out is:

(a) 3.5%

(b) 4.0%

(c) 5.0%

(d) 7.0%

B9. A futures contract has a tick value of \$25 per basis point. A trader holds 10 long contracts and the futures price drops by 30 basis points overnight. The overnight P&L is:

(a) −\$250

(b) −\$750

(c) −\$7,500

(d) −\$25,000

B10. Which of the following is *not* generally a reason to use derivatives for hedging?

(a) Lower transaction costs than cash markets.

(b) Higher liquidity at long maturities.

(c) Off-balance-sheet treatment.

(d) Guaranteed elimination of all risk.

Section C — Computational Problems ($5 \times 10 = 50$ pts)

C1 — Bond pricing and duration (10 pts)

A 3-year corporate bond pays a 6% *annual* coupon on a \$100 face value. The bond's yield-to-maturity is 5% (continuously compounded).

(a) Compute the bond's current price.

(b) Compute its Macaulay duration.

(c) Use duration to estimate the new price if the yield rises by 25 basis points (to 5.25%).

C2 — Forward pricing with a discrete dividend (10 pts)

A stock currently trades at \$60 and will pay a single \$1 dividend in 3 months. The risk-free rate is 4% (c.c.). Consider a 9-month forward on this stock.

(a) Compute the no-arbitrage 9-month forward price.

(b) The market quotes the 9-month forward at \$63. Identify an arbitrage strategy. Show the cash flows at $t = 0$, $t = 0.25$, and $t = 0.75$.

(c) Compute the risk-free profit today.

C3 — Margin account and daily settlement (10 pts)

A trader takes a *long* position of one KOSPI 200 futures contract on Day 0 at index level $F_0 = 400.0$. Contract multiplier: ₩250,000 per index point. Initial margin: ₩10,000,000. Maintenance margin: ₩7,500,000. Daily index closes:

Day	0	1	2	3	4
Index close	400.0	405.0	398.0	388.0	392.0

(a) Compute the daily mark-to-market gain/loss for each of Days 1–4.

(b) Compute the end-of-day margin-account balance for each day, before any required deposit.

(c) Identify the day a margin call is triggered, and compute the deposit required to restore the balance to the initial margin.

C4 — USD/KRW forward and arbitrage (10 pts)

Today, USD/KRW spot is ₩1,400 per USD, $r_{\text{KRW}} = 2.0\%$, $r_{\text{USD}} = 4.5\%$ (both c.c.).

(a) Compute the no-arbitrage 6-month USD/KRW forward rate.

(b) The market quotes the 6-month forward at ₩1,400 (equal to spot). Construct an arbitrage strategy (per 1 USD of forward notional) and compute the KRW profit.

(c) In one sentence, explain why $r_{\text{USD}} > r_{\text{KRW}}$ implies USD trades at a forward *discount* relative to spot.

C5 — Valuing a live forward (10 pts)

Six months ago, an investor entered a 1-year *long* forward on a non-dividend-paying stock. At that time the stock was \$40 and the risk-free rate was 5% (c.c.), so the forward was struck at the fair price $F_0 = 40 \cdot e^{0.05}$.

(a) Compute F_0.

(b) Today, with 6 months remaining, the stock has risen to \$45 and r is still 5%. Compute the current 6-month forward price F_t.

(c) Compute the value f of the original long position.